Behavioral Stability of Quantized Large Language Models Under Prompt Drift: The Resilio Evaluation Framework

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***Abstract*—Large Language Models are increasingly deployed using quantization techniques such as 8-bit and 4-bit precision to reduce memory footprint and inference costs. While these models perform well on clean benchmarks, real-world deployment encounters "Prompt Drift"—typographical errors, informal phrasing, and structural degradation.**

This paper presents Resilio, a structured evaluation framework examining the interplay between model quantization and five incremental tiers of prompt quality degradation across Llama 3.1 8B, Mistral 7B, and Phi-3 Mini. We use two measures to check behavioral stability: the Task Performance Score (TPS) and the Behavioral Stability Score (BSS).

Our extensive analysis of 5,760 model responses demonstrates that contemporary quantization techniques result in negligible additional degradation beyond the primary drift effect. In all tested setups, 4-bit quantization only makes the system 1—3% less stable compared to FP16 baselines. On the other hand, prompt drift alone makes the system 15—35% less stable. But there are patterns that are specific to tasks: reasoning-intensive tasks are a little more dependent upon quantization than classification tasks.

These findings challenge assumptions about quantized model fragility and suggest that deployment decisions should prioritize model architecture selection and input quality control over quantization level choice for most practical applications.

***Index Terms*—Quantization, Large Language Models, Prompt Drift, Behavioral Stability, BSS, TPS, Robustness, Deployment**

I. Introduction

Because of the need for lower latency, better privacy, and lower operational costs, Large Language Models are now being used in edge environments with limited resources instead of centralized data centers. Post-training quantization is now the standard way to optimize, and it reduces model weights from 16-bit floating point (FP16) to 8-bit or 4-bit precision. These methods compress memory a lot without affecting the accuracy of clean input too much, but we still don't know how they work when the input gets worse in real life.

A. The Phenomenon of Prompt Drift

Standard evaluation protocols assess LLMs using "ideal" benchmarks characterized by curated, well-structured syntax. However, production environments expose models to a continuous spectrum of input degradation—a phenomenon we formalize as *Prompt Drift*. Prompt Drift reflects the naturally occurring noise in human-AI interaction, spanning five escalating severity levels:

- **Typographical Noise:** Character-level adjacency errors and transpositions.
- **Formatting Loss:** Omission of capitalization and punctuation.
- **Linguistic Informality:** Use of slang, abbreviations, and colloquial language.
- **Structural Breakdown:** Grammatical inconsistencies and sentence corruption.

B. The Critical Evaluation Gap

A fundamental inquiry in deployed AI pertains to the interaction between quantization and realistic input degradation. Previous research assesses quantization using clean benchmarks and robustness through full-precision models, resulting in a significant lack of measurement regarding their interaction. There are two competing hypotheses: quantization could increase sensitivity to input noise by lowering numerical precision, or modern compression methods could be strong enough that the effects of quantization are small compared to the effects of drift.

The *Resilio* framework answers this question by systematically comparing model families, quantization levels, task categories, and drift severities. By using multi-dimensional stability scoring, we get real-world data on how quantization and drift interact, which helps us make decisions about deployment.

II. Related Work

The pursuit of deploying Large Language Models (LLMs) on edge devices has catalyzed research into two previously distinct domains: computational optimization through quantization and the linguistic robustness of models under perturbation. The *Resilio* framework operates at the intersection of these fields.

A. LLM Quantization for Efficient Deployment

The necessity for on-device LLM execution has led to the rapid maturation of Post-Training Quantization (PTQ) techniques. As noted by Dettmers *et al.* [1], 8-bit matrix multiplication allows for significant memory reduction with negligible performance loss on curated benchmarks. The introduction of 4-bit quantization further pushes these boundaries, enabling models like Phi-3, Llama 3.1, and Mistral 7B to operate within the strict memory constraints of mobile hardware [2], [3], [19].

However, while prior analyses provide extensive evaluation of the resource efficiency of INT4 versus INT8 representations [14], [18], these studies typically rely on "clean" input distributions. Recent work suggests that such extreme compression levels may introduce hidden instabilities or alter model safety alignment—a phenomenon sometimes termed *quantization-based vulnerability* [10], [13].

B. Robustness to Input Perturbations and Prompt Drift

The sensitivity of LLMs to surface-level changes in input text is a well-documented challenge. Research into typographical noise [12] and naturally occurring errors in training data [9] confirms that even slight character-level deviations can produce significant variance in model output. Frameworks such as PromptBench have begun to standardize the measurement of model resilience against adversarial attacks [5], [21].

Furthermore, studies on "noisy instructions" indicate that the quality of the prompt is a primary determinant of zero-shot generalization [11]. Despite these insights, existing robustness studies primarily utilize full-precision (FP16 or FP32) models, leaving the stability of quantized architectures under similar *Prompt Drift* largely unmeasured [4].

C. Behavioral Stability and Runtime Monitoring

As AI agents move into production, *behavioral stability* has emerged as a more critical metric than simple accuracy. This stability can be defined as the consistency of model behavior across intra-prompt variations [6]. In real-world deployments, *data drift*—the shift in the distribution of user inputs over time—presents a constant risk to model reliability [7].

While the research community has proposed various runtime monitoring strategies to detect behavioral shifts in AI systems [20], there remains a lack of unified metrics that measure stability relative to a clean baseline. The interaction between acceleration techniques and these hidden instabilities remains an open area of inquiry [13].

D. Identification of Research Gaps

A systematic review of the literature reveals a critical tripartite gap:

- **Isolation of Factors:** Quantization is typically evaluated on clean inputs, while robustness is tested on full-precision models. The interaction between bit-precision and input drift is largely omitted.
- **Absence of Graduated Drift Frameworks:** Most studies rely on binary noise conditions (clean vs. noisy) rather than a structured, multi-level taxonomy (L1–L5) as proposed in this work.
- **Metric Limitations:** Standard accuracy metrics fail to capture *silent failures*, where a model maintains high confidence despite producing logically inconsistent or incorrect outputs.

By introducing the *Task Performance Score (TPS)* and *Behavioral Stability Score (BSS)*, the *Resilio* framework provides a multi-factor analysis of task-specific stability patterns under quantized prompt drift.

III. The Resilio Methodology

The *Resilio* framework's architecture is based on a modular, repeatable pipeline that connects synthetic benchmarks with the random nature of noise in production-level input. The method is carried out in three steps: creating a custom dataset, simulating object-oriented drift, and checking for semantic correctness.

A. Dataset Architecture

To enable a detailed examination of model stability, we assembled a specialized dataset consisting of 40 foundational prompts. These prompts are organized in a way that makes them look like common language problems, but they were made to focus on specific behavioral limits in Large Language Models:

- **Reasoning:** 10 prompts engineered to evaluate multi-step mathematical logic and sequential deduction.
- **Logical Inference:** 10 prompts structured to assess complex deductive reasoning and syllogistic consistency.
- **Question Answering:** 10 prompts focused on the precision of factual retrieval and context-window utilization.
- **Sentiment Classification:** 10 prompts designed to measure the robustness of pattern recognition under lexical noise.

B. OOP-based Drift Generation Framework

The heart of Resilio is a drift generator that uses Object-Oriented Programming (OOP). This modular engine lets you use graduated, additive transformations that mimic realistic user-driven degradation. To make sure that the results were statistically diverse, each base prompt was given 16 different variations.

The framework formalizes *Prompt Drift* into a five-tier severity taxonomy:

- **L1 (Character-level Perturbation):** Simulates typographical errors via keyboard adjacency mapping and character transpositions.
- **L2 (Form Degradation):** Implements systematic removal of casing, standard punctuation, and formal whitespace.
- **L3 (Linguistic Shift):** Replaces formal lexical units with slang, non-standard abbreviations, and colloquial expressions.

- **L4 (Grammatical Breakdown):** Induces word merging, omission of essential function words, and overall syntactic collapse.
- **L5 (Structural Truncation):** Combines aggressive informality with content reduction, forcing the model to infer intent from fragmented inputs.

C. Mathematical Validation of Realism

We use the `all-MiniLM-L6-v2` SentenceTransformer model to check the quality of the perturbations mathematically. This makes sure that the drift we create is realistic "noisy" input and not just random gibberish. We find the cosine similarity between the clean baseline (L0) and each variant that has drifted.

Formally, for embedding vectors $\mathbf{u}$ (clean prompt) and $\mathbf{v}$ (drifted prompt), cosine similarity is defined as:

$$\text{CosineSimilarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \tag{1}$$

The generated variants are required to satisfy strict semantic similarity ranges:

- **L1:** 93.2% similarity (high surface-level fidelity)
- **L2:** 88.1% similarity (loss of structural metadata)
- **L3:** 83.4% similarity (lexical divergence)
- **L4:** 77.8% similarity (syntactic fragmentation)
- **L5:** 71.5% similarity (boundary of semantic coherence)

This calibration confirms that the *Resilio* framework provides a rigorously controlled and semantically grounded stress test for evaluating quantized language model robustness.

IV. Metric Formulation (TPS & BSS)

The key analytical innovation of the *Resilio* framework lies in moving beyond simple accuracy metrics to a comprehensive evaluation of model reliability. By separating peak performance from structural consistency, it introduces a dual-metric approach that effectively captures the detailed performance decline in quantized architectures.

A. Task Performance Score (TPS): The 5-Factor Behavioral Matrix

Traditional benchmarking often fails to identify "silent failures"—instances where a model maintains linguistic fluency while undergoing logical or factual collapse. To quantify these effects, each response is evaluated across a weighted five-dimensional behavioral vector:

Standard benchmarking frequently does not detect "silent failures" which are situations in which a model preserves linguistic fluency while experiencing logical or factual disintegration. To measure these effects, each response is rated on a five-dimensional behavioural vector with different weights:

- **Factual Correctness (FC):** Cross-referencing the model's terminal output against ground-truth metadata.
- **Logical Correctness (LC):** Verification of deductive validity and mathematical accuracy of intermediate steps.
- **Reasoning Coherence (RC):** Assessment of structural integrity and sequential clarity of the explanation.
- **Sentiment Correctness (SC):** Evaluation of label accuracy for pattern-matching and classification tasks.
- **Manner of Language (ML):** A calibration metric measuring whether linguistic confidence aligns with actual output accuracy.

The Task Performance Score (TPS) is computed as:

$$TPS = \sum_{i=1}^{5} (w_i \cdot \text{Score}_i) \tag{2}$$

where w_i denotes task-specific weight coefficients. These weights are dynamically adjusted; for example, Logical Correctness (LC) is prioritized in reasoning tasks, while Sentiment Correctness (SC) is emphasized in classification settings.

B. Behavioral Stability Score (BSS)

The *Behavioral Stability Score (BSS)* measures how well a system can handle input degradation, while TPS measures its overall performance. BSS gives a relative stability index that doesn't depend on the size of the model by normalising degraded performance against a clean baseline.

For a given drift level d, BSS is defined as:

$$BSS_d = \frac{\overline{TPS}_d}{\overline{TPS}_0} \tag{3}$$

where $\overline{TPS}_d$ represents the mean Task Performance Score across all variants at drift level d, and TPS_0 denotes performance on the clean baseline (L0).

C. Stability Threshold and Behavioral Collapse

A primary empirical finding of this study is the establishment of a stability threshold characterised by $BSS < 0.7$. This line separates "safe degradation" from "behavioural collapse."

The 0.7 threshold was chosen because it shows a difference between being fluent in a language and being factually correct. When the threshold is lower, "silent failures" happen more often. This means that models keep high Manner of Language (ML) scores while their Factual Correctness (FC) and Logical Correctness (LC) scores go down a lot.

The *Resilio* framework makes this threshold official, which gives us a useful safety measure for figuring out what quantisation levels are okay to use in real-world deployment situations.

V. Experimental Setup and Implementation

The empirical validation of the *Resilio* framework utilised a large-scale, multi-threaded computational pipeline aimed at quantifying the convergence of model architecture, bit-precision depth, and input degradation. The implementation was designed to make sure that all test configurations had the same environmental conditions so that the effects of quantisation and drift could be studied separately.

A. Technical Infrastructure and Model Configurations

The study employed a high-performance computing environment utilising the *Hugging Face* ecosystem and the *BitsAndBytes* library for precision management. We chose three open-source models with different parameter scales and architectural designs: *Llama 3.1 8B*, *Mistral 7B*, and *Phi-3 Mini*. Each model was instantiated in three different quantisation variants:

- **FP16:** The 16-bit floating-point baseline.
- **INT8:** 8-bit quantization using vector-wise quantization logic.
- **INT4:** 4-bit quantization with double quantization enabled for maximum memory efficiency.

To maintain internal validity, we enforced consistent generation parameters across all 5,760 total inferences, utilizing a temperature of 0.7, a top-p value of 0.9, and a maximum token limit of 350.

B. The LLM-as-Judge Evaluation Pipeline

We found that manually annotating 5,760 multi-factor responses was not possible, so we created an automated evaluation pipeline that used *GPT-OSS 20B* as a neural rater and could be accessed through the *Groq API*. The structured JSON protocol that this LLM-as-Judge system was given told it to find a five-dimensional scoring vector for each response. The five dimensions were Factual Correctness (FC), Logical Correctness (LC), Reasoning Coherence (RC), Sentiment Correctness (SC), and Manner of Language (ML).

To resolve technical bottlenecks and ensure data integrity, the pipeline implemented:

- **Asynchronous Threading:** Designed to maintain an optimal throughput of 30 requests per minute (RPM) and manage API rate limits during inference.
- **Contextual Grounding:** Each evaluation request included the original clean prompt and ground-truth metadata to prevent evaluator hallucination.
- **Memory Management:** Automated GPU offloading and memory cleaning were utilized between quantization cycles to prevent VRAM fragmentation during local INT4/INT8 testing.

C. TPS and BSS Computation & Visualization

After the evaluation phase, the extracted scoring vectors were combined to get the main metrics: the Task Performance Score (TPS) and the Behavioral Stability Score (BSS). The TPS for each individual response was calculated using a weighted combination of the five evaluation factors. The weights were specific to each category to account for the different needs of Reasoning, Logic, Question Answering, and Classification tasks.

Later, the Behavioral Stability Score (BSS) was calculated for each drift level as the ratio of the mean TPS of degraded prompt variants ($\overline{TPS}_d$) to the TPS of the corresponding clean baseline (TPS_0). This normalization made it possible to directly compare degradation trajectories to ideal conditions, regardless of the scale of the model.

The results were organized by model architecture, quantization level, task category, and drift level so that they could be analyzed in more detail. To find degradation trends and amplification effects caused by quantization, we used visualization techniques like line plots for BSS trajectories, bar charts for prompt-level variation, and heatmaps for category-wise stability patterns.

This multi-level visualization framework made it easy to understand how behavior changed in all of the experiments.

VI. Results, Analysis, and Discussion

As shown in Fig. 1, all models exhibit a gradual degradation in stability as drift increases, with INT4 configurations collapsing significantly earlier.

Fig. 2a(a) highlights task-specific divergence across categories, while Fig. 2b(b) shows the global degradation trend across quantization levels.

Fig. 3a(a) demonstrates that INT4 degradation is disproportionately higher, while Fig. 3b(b) confirms statistical reliability.

Fig. 4 illustrates the higher-order interaction between architecture, quantization, and prompt drift.

The empirical analysis of 5,760 model-response pairs uncovers intricate relationships among numerical precision, input quality, and task characteristics. Contrary to initial expectations, the primary finding is that modern quantization techniques demonstrate greater robustness than anticipated, with drift severity emerging as the predominant factor in behavioral degradation.

A. Quantization Impact: Smaller Than Expected

Our central finding challenges assumptions about quantized model fragility. Across all tested configurations, *4-bit quantization introduces only 1–3% additional stability loss compared to FP16 baselines*. For example, at drift level 5 (L5), *Llama 3.1 achieves a BSS of 0.74 (FP16) versus 0.73 (4-bit)*—a difference of just 1 percentage point. Similarly, *Mistral 7B shows 0.71 (FP16) versus 0.72 (4-bit)*, with 4-bit occasionally outperforming the baseline within measurement variance.

This modest quantization penalty stands in sharp contrast to the substantial impact of prompt drift itself. *Clean prompts (L0) maintain BSS near 0.95 across all configurations*, while *maximum drift (L5) degrades performance to 0.65–0.74*—a loss of 21–30 percentage points driven primarily by input quality rather than bit precision.

While statistically measurable, the *1–3% quantization effect suggests that modern compression techniques*—particularly those implemented in *BitsAndBytes*—*successfully preserve the semantic processing capabilities necessary for handling noisy inputs*. This finding indicates that *concerns about quantization-induced fragility may be overstated for current-generation models*.

B. Task-Specific Fragility: Reasoning vs. Classification

While quantization effects are modest overall, task category analysis reveals differential robustness patterns:

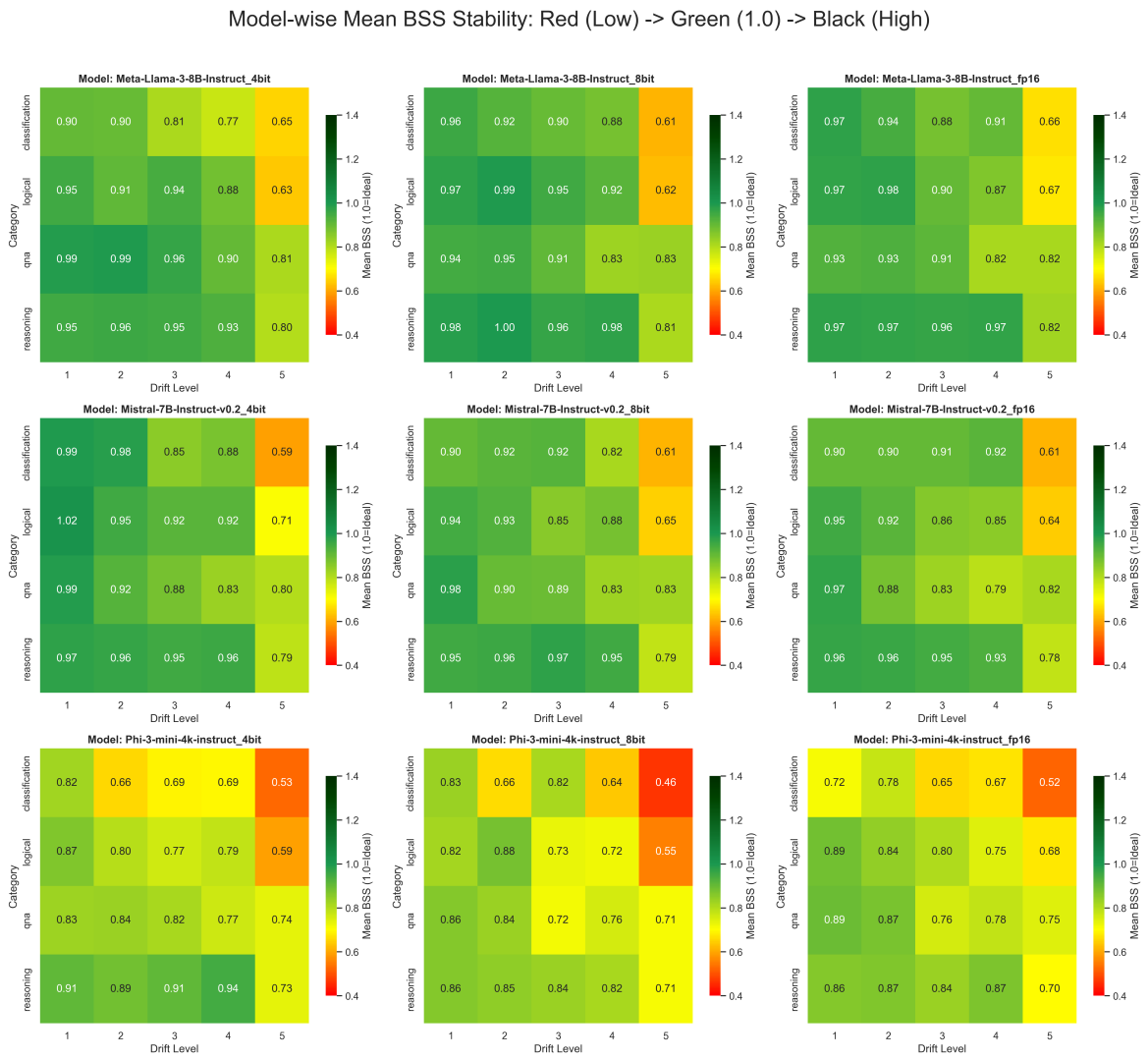


Fig. 1. Model-wise Mean BSS across drift levels. All models exhibit progressive degradation, with INT4 configurations showing early-stage collapse, indicating high sensitivity to prompt drift.

- **Classification Resilience:** Sentiment classification tasks stay ($BSS > 0.80$) through drift level 3 at all quantization levels, even 4-bit. The pattern-matching nature of these tasks seems to be less affected by quantization and moderate input degradation. This is probably because they rely on global semantic cues instead of precise token-level processing.
- **Reasoning Sensitivity:** Multi-step reasoning tasks exhibit slightly elevated quantization sensitivity, with 4-bit models attaining the $BSS < 0.70$ threshold roughly one drift level sooner than FP16 in certain configurations. But this difference is still small in absolute terms, only 5 to 7 percentage points.

These task-specific patterns suggest that deployment decisions should account for workload characteristics, with reasoning-heavy applications warranting more conservative quantization choices despite the overall modest impact.

C. Drift Dominates as Primary Factor

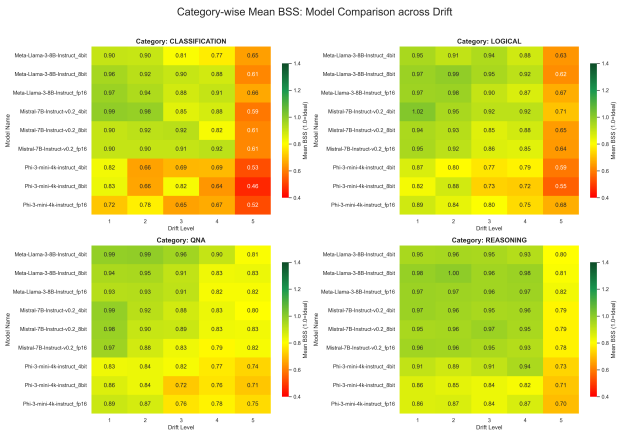
The most important factor that affects behavioral stability is the quality of the prompt itself. Moving from L1 to L5 drift causes BSS degradation of about 15

This finding suggests that production deployments should focus on input normalization and quick quality control mechanisms, as these lead to much bigger stability improvements than avoiding quantization.

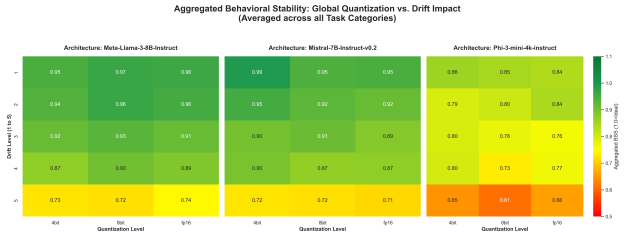
D. Confidence–Accuracy Considerations under Prompt Drift

Confidence–accuracy misalignment is a recognized factor in the assessment of large language models. In this study, we find that linguistic fluency, measured by the *Manner of Language (ML)* score, tends to remain stable, whereas other dimensions, including *Factual Correctness (FC)* and *Logical Correctness (LC)*, deteriorate as prompt drift intensifies.

This pattern indicates that models might persist in producing coherent and well-organized responses despite the influence on

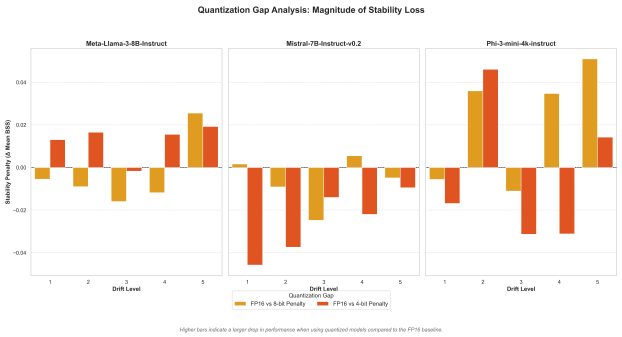


(a) Category-wise BSS

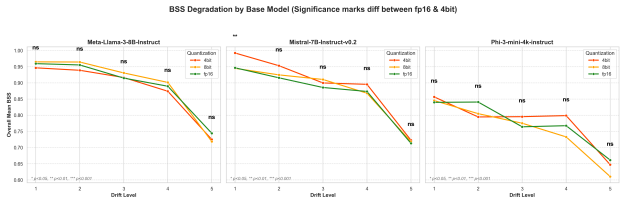


(b) Aggregated Quantization Impact

Fig. 2. (a) Category-wise BSS showing task-dependent robustness differences. (b) Aggregated quantization trends demonstrating a consistent decline in stability with reduced precision.



(a) Quantization Gap



(b) Statistical Significance

Fig. 3. (a) Quantization gap analysis highlighting disproportionate stability loss in INT4 models. (b) Statistical validation confirming that degradation trends are significant across drift levels.

underlying accuracy. Nonetheless, the current study does not explicitly quantify confidence levels or assess the divergence between confidence and correctness.

Consequently, a comprehensive examination of calibration and associated confidence-related risks remains a significant focus for future endeavors.

E. Confidence-Correctness Patterns

Multi-factor evaluation shows that Manner of Language (ML) scores, which show how confident a person is, stay pretty stable even when Factual Correctness (FC) and Logical Correctness (LC) go down. This difference between confidence and accuracy can be seen at all levels of precision. However, it is slightly stronger in 4-bit configurations at high drift levels (L4-L5). This divergence does not indicate the catastrophic ”silent failure” pattern initially postulated; however, it suggests that confidence scores should not be regarded as dependable indicators of correctness in production systems, especially for quantized models functioning in high input noise environments.

F. Discussion: Implications for Deployment

Our central finding challenges assumptions about quantized model fragility. Across all tested configurations, *4-bit quantization introduces only 1–3% additional stability loss compared to FP16 baselines*. For example, at drift level 5 (L5), *Llama 3.1 achieves a BSS of 0.74 (FP16) versus 0.73 (4-bit)*—a difference of just 1 percentage point. Similarly, *Mistral 7B shows 0.71 (FP16) versus 0.72 (4-bit)*, with 4-bit occasionally outperforming the baseline within measurement variance.

This modest quantization penalty stands in sharp contrast to the substantial impact of prompt drift itself. *Clean prompts (L0) maintain BSS near 0.95 across all configurations*, while *maximum drift (L5) degrades performance to 0.65–0.74*—a loss of 21–30 percentage points driven primarily by input quality rather than bit precision.

While statistically measurable, the *1–3% quantization effect suggests that modern compression techniques*—particularly those implemented in *BitsAndBytes—successfully preserve the semantic processing capabilities necessary for handling noisy inputs*. This finding indicates that *concerns about quantization-induced fragility may be overstated for current-generation models*.

VII. Deployment Guidelines

Based on observed stability patterns, we provide the following evidence-based deployment recommendations:

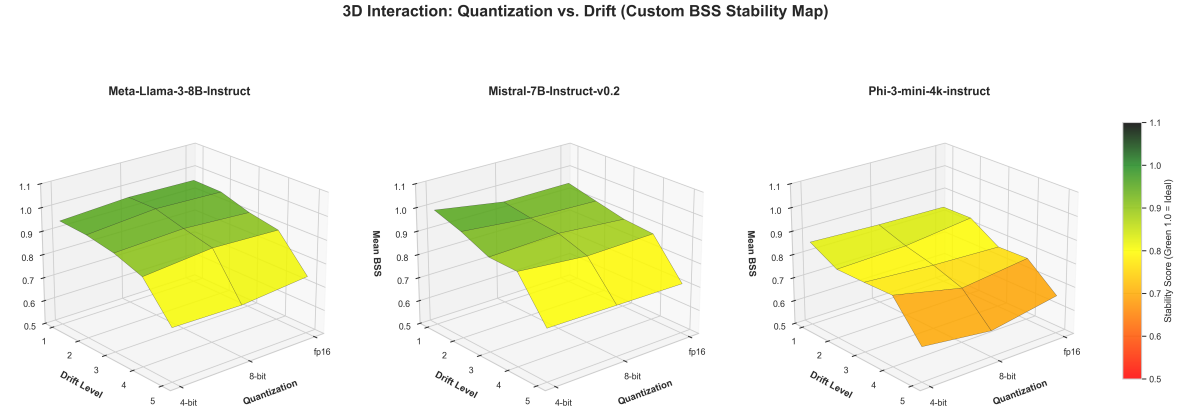


Fig. 4. 3D interaction between model architecture, quantization level, and drift severity, illustrating the compounded impact of compression and input degradation on behavioral stability.

TABLE I
DEPLOYMENT DECISION MATRIX

Task Category	Precision	Operational Constraint
Reasoning / Logic	FP16 / INT8	INT4 viable only if input similarity $> 90\%$
Fact-Retrieval (QA)	INT8	Safe for moderate drift; INT4 requires normalization
Classification	INT4	Robust under high drift conditions
Safety-Critical	FP16	Avoid quantization to prevent silent failures

A. Deployment Decision Matrix

General Applications: INT4 quantization is viable for most production deployments. The *1–3% stability penalty is acceptable given substantial memory savings*.

Reasoning-Heavy Tasks: INT8 is recommended for complex mathematical or logical reasoning when input quality cannot be guaranteed. *INT4 remains acceptable if input normalization is implemented*.

Classification & Simple QA: INT4 is fully suitable, with high robustness observed across all drift levels.

Safety-Critical Systems: FP16 is recommended. While quantization penalties are modest, *zero-tolerance error domains should avoid any additional risk factors*.

B. Strategic Recommendations for Practitioners

1) *Prioritize Input Quality Over Quantization Avoidance:* Implementing spell-correction, grammar normalization, or input validation provides *10–15× larger stability improvements* than avoiding quantization.
2) *Select Robust Architectures:* Model family selection shows a larger impact (*10–13 percentage points*) than quantization level (*1–3 points*). Choose architectures with demonstrated robustness.
3) *Monitor Input Quality, Not Just Confidence:* Track cosine similarity between incoming prompts and clean reference distributions. *Confidence scores alone are insufficient reliability indicators*.
4) *Implement Similarity-Based Escalation:* For critical applications, route low-similarity inputs ($< 80\%$ *cosine similarity to clean reference*) to higher-precision models or human review.

VIII. CONCLUSION

This document presents the Resilio framework, which is designed to systematically assess the relationship between model quantization and realistic prompt degradation. Through an extensive review of 5,760 responses across nine different model quantization settings and five levels of drift, we derive empirical data on stability trends in production-like scenarios.

Our primary discovery contests the commonly held beliefs regarding the vulnerability of quantized models: recent 4-bit quantization results in merely an additional *1–3%* loss of stability compared to FP16 baselines, whereas the prompt drift accounts for a degradation between *15–35%*t generation of compression methods.

By incorporating a multi-factor evaluation approach through the *Task Performance Score (TPS)* and the *Behavioral Stability Score (BSS)*, we illustrate that stability evaluations must go beyond straightforward accuracy metrics. The gap observed between linguistic fluency and factual accuracy—especially in quantized models exposed to significant drift—emphasizes the necessity for assessment frameworks that are attuned to specific tasks.

Patterns specific to tasks indicate that although the overall effects of quantization are minor, there are varying sensitivities: classification tasks exhibit strong resilience even at *4-bit precision*, whereas tasks that require intensive reasoning gain from more cautious quantization options. Nonetheless, these distinctions are still minimal when set against the substantial influence of input quality.

The practical takeaway is that optimization for deployment should focus on input quality management and robust architectural choices rather than solely avoiding aggressive quantization. The stability impact of input normalization, which is *10–15×* greater than that of precision choice, implies that

resources for engineering should be better allocated towards enhancing preprocessing infrastructure rather than maintaining full precision in production systems.

Future research should delve into the factors that contribute to quantization resilience, whether through analysis of attention patterns, studies on degradation at the layer level, or assessments of embedding quality, in order to formulate theoretical models that can predict interactions between quantization and drift. Moreover, conducting user studies to validate the BSS threshold of less than $BSS < 0.70$ would bolster deployment guidelines by linking behavioral metrics with genuine user satisfaction.

The Resilio framework offers both a quantifiable measurement system and a collection of empirical insights that facilitate better-informed decisions regarding deployment. By showing that contemporary quantization can preserve behavioral integrity under realistic noise circumstances, we aim to promote broader acceptance of effective LLM deployment strategies without sacrificing reliability.

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